ResearchGate

See discussions, stats, and author profiles for this publication at: https://www.researchgate.net/publication/262528520

Analytical Prediction of Melt Depth Inside Ti-6Al-4V During Laser Materials Processing

Conference Paper · November 2012

DOI: 10.1115/IMECE2012-89883

CITATIONS
0

READS
285

2 authors:

Dineo Ngwenya
Fhatani Consulting Engineers
1 PUBLICATION **0** CITATIONS

SEE PROFILE

Franz-Josef Kahlen
University of Cape Town
35 PUBLICATIONS **483** CITATIONS

SEE PROFILE

All content following this page was uploaded by Franz-Josef Kahlen on 23 May 2014.

The user has requested enhancement of the downloaded file.

Proceedings of the ASME 2012 International Mechanical Engineering Congress and Exposition
IMECE2012
November 9-15, 2012, Houston, Texas, USA

IMECE2012-89883

ANALYTICAL PREDICTION OF MELT DEPTH INSIDE TI-6AL-4V DURING LASER MATERIALS PROCESSING

D. Ngwenya
Department of Mechanical Engineering,
University of Cape Town
Cape Town
South Africa

F.J. Kahlen
fj.Kahlen@uct.ac.za
Department of Mechanical Engineering
University of Cape Town
Cape Town
South Africa

ABSTRACT

The prediction of melt depth evolution inside Ti-6A-4V during laser-material processing enables the understanding of how heat affects the microstructure and phase transformations. However, modeling time increases with increased complication of mathematical functions. This work investigates the feasibility of approximating a melt depth profile which features the inverse of a complimentary error function (erfcinv) with a simpler mathematical function. It is proposed that for the ease of calculations, a melt depth profile containing the natural log function (ln) be used to approximate an erfcinv-bearing melt-depth-predicting equation. An introduction of a ratio R is made as a correction factor. This ratio is then applied to the natural-log-function-bearing melt-depth-predicting equation thus improving upon its approximation. In addition, the dependency of R on the affective laser parameters such as pulse duration and incident power density is investigated.

INTRODUCTION

Titanium and its alloys are used in applications where their most recognized properties: excellent combination of strength and toughness, corrosion resistance and high melting point, justify their selection. Nevertheless, the application of titanium alloys under severe wear and friction conditions is highly restricted due to their poor tribology properties [1].

Laser technology is used extensively in the advancement of manufacturing techniques. It is significantly appreciated as a machining tool in the surface modification of titanium alloys [1-5]. Pleshakov et al [5] conducted an experimental study on laser surface modification of Ti-6Al-4V alloy with silicon carbide. They showed that laser modifying through alloying with elements of superior hardness properties, enhances the wear resistance of Ti-6Al-4V. The aim of improving surface properties was also achieved by Saleh et al [4] in their study of surface carburizing of Ti-6Al-4V alloy by laser melting. Sun et al [3] improved the micro-hardness of the Ti-6Al-4V surface and concluded that an increase in laser energy resulted in an increase in micro-hardness in clad zones. A study of surface modifications of Ti-6Al-4V by laser shock processing revealed that, for biomedical applications where roughening of the surface is required, laser shock processing increases the surface roughness [6].

The efficiency of laser-assisted processes is mostly controlled by the thermal history of the thermally affected areas of a material. The temperature profiles, melt zones, the time during which the material remains molten, and the solidification rates constitute a thermal history of a material [4]. An understanding of the temporal distribution of temperature during laser-material interaction is thus important. The prediction of characteristic changes such as phase transformations during rapid cooling and the final microstructure, require a thermal history [7]. A thermal history is influenced by the interaction of laser parameters and thermal properties of a material.

During surface irradiation of a material, only a portion of the laser energy is absorbed [7]. If this absorbed energy is high enough, melting will be inevitable. The study of the temporal evolution of temperature fields in a material during laser-induced heating and melting has attracted great interest [7-12]. The results obtained from such studies are important in

Copyright © 2012 by ASME

optimizing laser-assisted manufacturing processes, and identifying favorable processing parameters.

NOMENCLATURE

A	absorption coefficient
I	laser beam intensity (W/m^2)
L	latent heat of fusion (J/kg)
P	power (W)
R	ratio
T	temperature (K)
X	melt depth (m)
k	conductivity (W/mK)
t	irradiation time (s)
z	length (m)
α	thermal diffusivity (m^2/s)
ρ	density (kg/m^3)

Subscripts

0	initial state
l	liquid
s	solid

Mathematical model

To simplify the heat transfer problem, it is treated as a one-dimensional heat transfer problem. A schematic of the heat transfer problem is shown in Fig. 1.

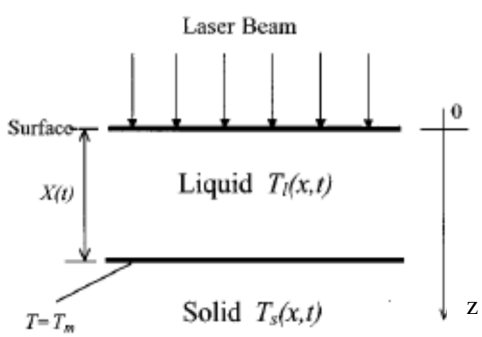

Figure 1: A semi-infinite slab under laser irradiation [8].

Furthermore, it is assumed that the diameter of the laser beam is broad enough when compared to the molten region on the surface. The thickness of the material is assumed to be much greater than the thermal penetration depth, so that a semi-infinite model can be accepted [9]. The thermo-physical properties of Ti-6Al-4V are assumed to be independent of temperature therefore; the density, thermal conductivity and specific heat capacity remain temperature-unaffected. Convection and radiation energy losses are ignored in this model. Therefore, the heat transfer is assumed to only be conductive. The entire laser-induced heating and melting in the material is divided into two temperature fields: before (T_s) and after melting (T_l). The thermal conduction model is described by the following boundary conditions:

$$\frac{\partial^2 T_l(z,t)}{\partial z^2} - \frac{1}{\alpha_l}\frac{\partial T_l(z,t)}{\partial t} = 0, \qquad 0 \le z < X \tag{1}$$

$$\frac{\delta^2 T_s(z,t)}{\partial z^2} - \frac{1}{\alpha_s}\frac{\partial T_s(z,t)}{\partial t} = 0, \qquad X \le z < \infty \tag{2}$$

$$T_l(z,t) = T_s(z,t) = T_m, \qquad z = X \tag{3}$$

$$k_l\frac{\partial T_l(z,t)}{\partial z} = k_s\frac{\partial T_s(z,t)}{\partial z} - \rho L\frac{dX}{dt}, \quad z = X \tag{4}$$

$$-k_l\frac{\partial T_l(z,t)}{\partial z} = AI, \qquad z = 0 \tag{5}$$

$$T_s(z,t) = T_0\,, \qquad z \to \infty \tag{6}$$

An analytical solution to the laser beam-targeted material heat transfer is developed from the boundary conditions described by Eqs. (1) - (6). The solid phase temperature profile is developed to be:

$$T_s = T_{initial} + \frac{P}{2\pi\sqrt{\alpha_s t}k_s} erfc\left(\frac{z}{\sqrt{4\alpha_s t}}\right) \tag{7}$$

Satisfying Eq. (5), the liquid phase temperature profile is obtainable by assuming that it is in the form:

$$T_l = \frac{-AI}{k_l}z + Bz^2 + C \tag{8}$$

By using $T_s = T_m$ in Eq. (3), the equation for melt depth is obtained as:

Copyright © 2012 by ASME

$$X = 2\sqrt{\alpha_s t}\, InverseErfc\left[\frac{-2k_s\pi\sqrt{\alpha_s t}(T_i - T_m)}{P}\right] \quad (9)$$

Using the formulae for X, T_s and T_l in Eq. (4) yields:

$$B = -\frac{exp\left[\frac{-X^2}{4\alpha_s t}\right]P}{4X\alpha_s k_l \pi^{3/2} t} - \rho L\frac{dX}{dt}\frac{1}{2Xk_l} + \frac{AI}{2Xk_l} \quad (10)$$

And C is found by using $T_l = T_m$ in Eq. (3). Therefore:

$$T_l = \frac{-AI}{k_l}(z - X) + B(z^2 - X^2) + T_m \quad (11)$$

The thermo-physical properties of Ti-6Al-4V and its absorption coefficients at 1.06μm wavelength, are listed in Table 1 [9,13,14].

	Ti-6Al-4V
ρ (kg m^{-3})	4510
L (10^5 J kg^{-1})	2.9
T_m (K)	1928
k_s (Wm^{-1}K^{-1})	20[a]
α_s (10^{-6} m^2 s^{-1})	5.8[a]
k_l (Wm^{-1} K^{-1})	29[b]
α_l (10^{-6} m^2 s^{-1})	5.95[b]
A	0.433

Table 1: Table of thermo-physical properties of Ti-6Al-4V. [a]Data at 1500K and [b]data at T_m.

RESULTS AND DISCUSSION

The results are obtained using Eq. (7), (9), (11) and the thermo-physical properties listed in Table 1.

Laser melting

Analytical models for laser melting of titanium [8,9] showed that the growth rate of the melt depth is high at the beginning of laser irradiation and then low after a certain irradiation time. Equation (9) shows a similar trend as shown in Fig. 2. This behavior could be attributed to the idea that, at the beginning of the melting process the rate of laser energy absorbed is greater than that of conduction. This then allows for the absorbed energy to raise local temperatures beyond the melting point. Therefore, the melt depth increases rapidly until the earlier absorbed laser energy is less than or equal to the amount of energy being conducted. It follows that if there is more energy being conducted further into the work-piece than there is being absorbed, there will be less energy available to significantly raise local temperatures. According to Abd El-Ghany [11] the melt depth continues to increase to its maximum point where the absorbed laser energy is equal to the sum of the losses resulting from cooling and conduction in the liquid.

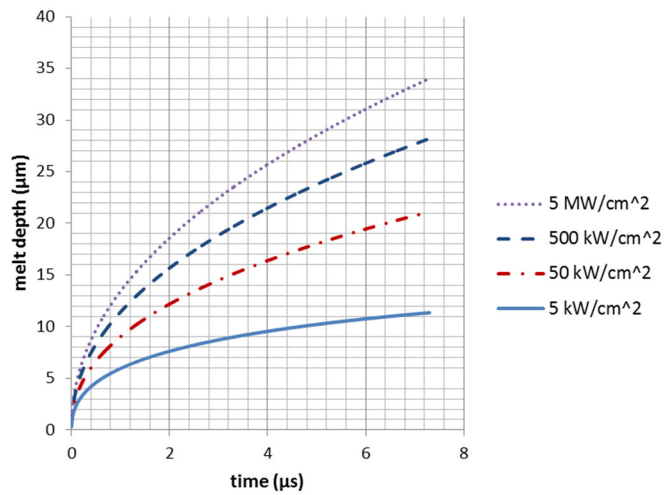

Figure 2: Variation of melt depth with irradiation time and laser beam intensity.

Effect of laser beam intensity on melt depth

From Fig. 2 it is evidenced that an increase in laser beam intensity results in an increase of melt depth. This trend is due to an increased availability of energy to raise temperatures to above the melting point. It is also shown in Fig. 2 that the melt depth increases rapidly with increasing irradiation time when the laser beam intensity is high, and increases slowly at lower laser beam intensities.

The variation of the calculated melt depth with varying laser power for an irradiation time of 0.13s is plotted in Fig. 3. The melt depth increases rapidly with increasing laser power when the laser power is low, and increases slowly when the laser power is high [8,9]. Such a trend is also observed in experimental investigations of laser welding Ti-6Al-4V of which the results are also shown in Fig. 3.

A comparison of the calculated melt depth and the experimental data in Fig. 3 shows a fairly good agreement. A significant difference in the two data sets is observed at large laser powers where unexpectedly, the theoretical prediction is an underestimate of the experimental data. Apart from the effects of the assumptions made in this article, on the accuracy

Copyright © 2012 by ASME

of the theoretical results, the effects of ignoring laser plasma and vapour result in the expectation of the theoretical results overestimating the melt depth. Vapour and laser plasma block the incident laser and absorb a portion of its energy [8]. Furthermore, a study of the nonlinear effects of laser-plasma interaction on melt surface temperature[16] showed that laser plasma increases in height and radius as laser power is increased [15]. The overall effect is that with an increase in power, there is a decrease in the total amount of energy available for absorption by the target surface, thus causing a decrease in melt depths. Therefore, the theoretical model in Fig. 3 should be an overestimate of the experimental data at high laser powers. It is appreciated that in the case of laser welding, the laser beam moves relative to the target surface. However, this article assumes that the laser beam and the target are stationary. This difference can be accounted for by applying an appropriate irradiation time to the theoretical model. The appropriate irradiation time is obtainable through knowing the beam spot diameter and scanning speed where $t = {D_0}/{v_0}$ where D_0 and v_0 are the beam spot diameter and the scanning velocity respectively [8]. A better agreement between the model presented by Eq. (9) and the experimental results is thus possible if the effects of plasma and vapor are accounted for in the model.

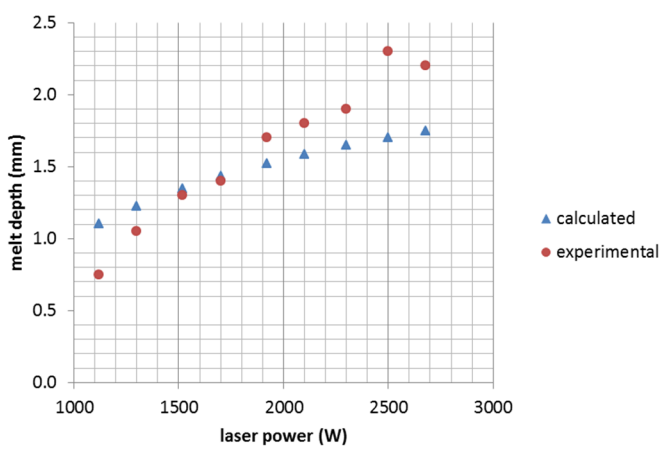

Figure 3: Comparison of the effect of laser power on melt depth during laser-Ti-6Al-4V surface interaction between experimental and calculated melt depth.

Effect of mathematical function approximation on melt depth

Numerical values of both erfc(x) and erfc^{-1}(x) are not as easily available as with logarithms [12]. This makes algebraic manipulation difficult. Therefore, the problem is avoided by attempting to approximate erfc(x) and erfc^{-1}(x) with simplified mathematical functions. Equation (12) is an approximation of Eq. (9).

$$X = -2\sqrt{\alpha_s t}\, ln\left[\frac{-2k_s\pi\sqrt{\alpha_s t}(T_i - T_m)}{P}\right] \tag{12}$$

A new multiplication factor R is introduced in Eq. (13) in order to improve the accuracy with which Eq. (12) approximates Eq. (9). It is the ratio of Eq. (9) to Eq. (12). Therefore

$$X = -2\sqrt{\alpha_s t}\, R \times ln\left[\frac{-2k_s\pi\sqrt{\alpha_s t}(T_i - T_m)}{P}\right] \tag{13}$$

As shown in Figs. 4-7, Eq. (12) is an overestimate of Eq. (9). However, the introduction of R and R_{mean} improves the accuracy of the approximation. The difference between R and R_{mean} is that the one is the average of the R values calculated for a specific beam intensity whilst the other is an average of values obtained for all modeled beam intensities (5 kW/cm^2, 50 kW/cm^2, 500 kW/cm^2 and 5 MW/cm^2) respectively. The values of R and R_{mean} are displayed in Table 3. Figure 5 shows that a better approximation of Eq. (9) is attainable with laser parameters of I = 50 kW/cm^2 and 0 < t < 8μs for R_{mean} 0.395. For I < 50 kW/cm^2, the use of R_{mean} leads to an underestimation of Eq. (9). Furthermore, for I > 5 kW/cm^2, the use of R_{mean} leads to an overestimation of Eq. (9). However, when only considering the use of R, the melt depth profile in Fig. 7 is an ideal approximation of Eq. (9). It is considered so because it has the lowest R value out of those displayed in Table 2. An ideal case is where the correction factor (R in this case) is kept as small as possible. A small correction factor indicates a relatively higher accuracy with which one equation approximates another. Therefore, within reasonable error, the approximating equation can be used without the inclusion of a correction factor.

Effect of laser intensity and irradiation time on R

The analytical modeling of R as a function of P showed that the value of R decreases with an increase in P values as shown in Figs. 8 and 9. For large incident powers, the ratio of X(erfc^{-1}) and X(ln) is minimized and fluctuates over a narrower range as compared to the ratio for smaller P values in Fig. 8. For large irradiation times, the ratio R reaches a plateau. Therefore, this suggests that the ratio of X(erfc^{-1}) to X(ln) eventually remains constant at large irradiation times. Therefore, the effect of incident laser beam intensity on R is that an increase in P values results in a decrease in R values.

Copyright © 2012 by ASME

Therefore, larger errors are associated with the approximation of Eq. (9) by Eq. (13) at smaller laser beam intensities. In addition, the effect of irradiation times on R is that R remains constant for large irradiation times. In Fig. 8, R exhibits a behavior that shows R decreasing rapidly at small incident laser beam intensities. It then decreases slowly at relatively larger laser beam intensities. The slow decrease of R values at larger laser beam intensities suggests that the accuracy of Eq. (13) in approximating the melt depth in Eq. (9) has a maximum. The desirable approximation is that which has the highest accuracy thus lowest R value. However, the lowest R value in this case is approximately 0.3. This implies that the proposed approximation is three times an overestimate of the predicted melt depth.

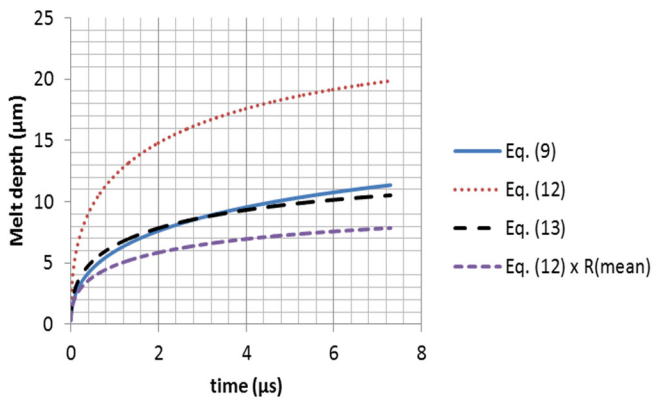

Figure 4: Melt depth against irradiation time for Ti-6Al-4V with I = 5 kW/cm^2.

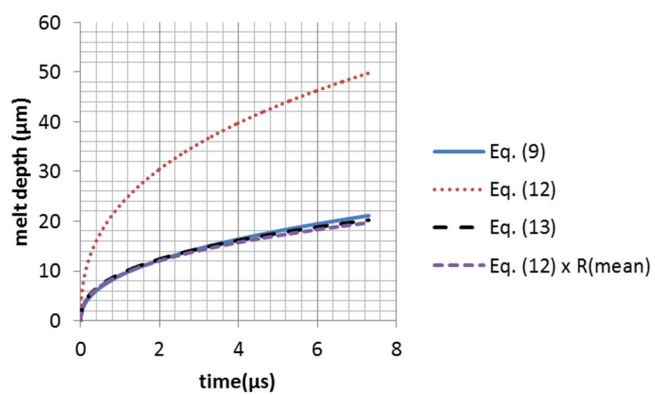

Figure 5: Melt depth against irradiation time for Ti-6Al-4V with I = 50 kW/cm^2

Table 2: Table of R and R_{mean} values. [a]An average of individual R values obtained for a constant I value and a range of t values. [b]An average of all individual R values.

I (kW/cm^2)	5	50	500	5000
[a]R	0.530	0.406	0.343	0.303
[b]R_{mean}	0.395			

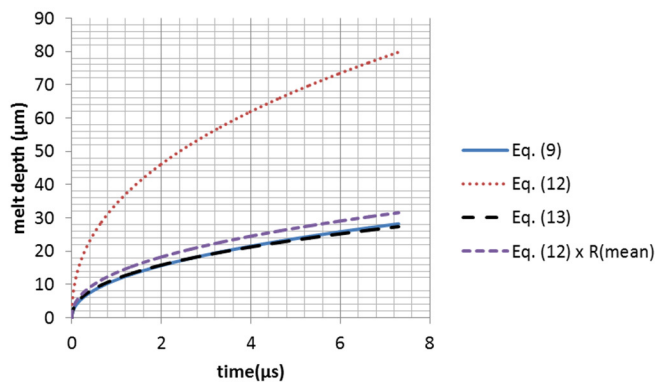

Figure 6: Melt depth against irradiation time for Ti-6Al-4V with I = 500 kW/cm^2.

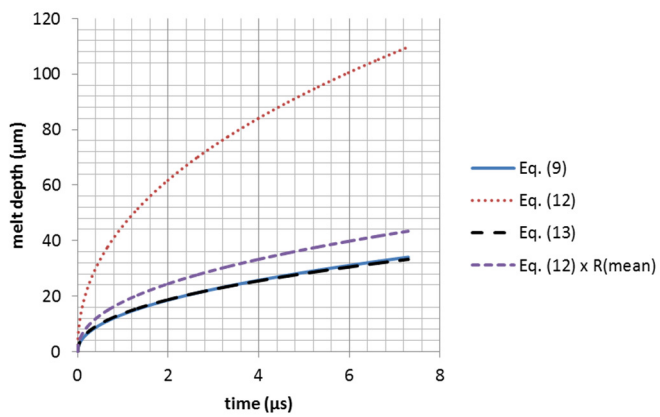

Figure 7: Melt depth against irradiation time for Ti-6Al-4V with I = 5 MW/cm^2.

Copyright © 2012 by ASME

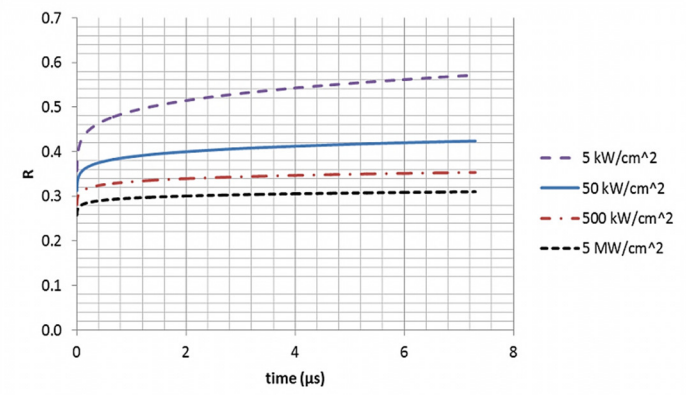

Figure 8: Variation of individual R values with irradiation times at different incident beam intensity.

CONCLUSIONS

An analytical method for developing a melt depth profile during laser heating and melting is presented. The analytical solution is applied for Ti-6Al-4V titanium alloy. The melt depth prediction showed a fair agreement with analytical predictions of other researches, more especially the experimental results. An attempt to approximate a melt depth equation that carries a complex mathematical function with a simpler one was made. A natural logarithmic function approximation of the inverse error function proved to be an overestimate. The use of a ratio R to account for the overestimate improved the approximation. For irradiation times $0 < t > 7.3$µs the R values applicable for intensities 5, 50, 500, 5000 kW/cm^2 were found to be 0.53, 0.406, 0.343 and 0.303 respectively. Although, for low intensities (5 kW/cm^2) the ratio R did not improve the approximation as well as it did for higher intensities (500 kW/cm^2 and 5 MW/cm^2). In a range of intensities modeled, the use of an average R value (R_{mean}) of 0.395 best improved the mathematical approximation with a laser intensity of 50 kW/cm^2. These findings are relevant to models that are designed to predict the melt depth evolution inside a material (especially Ti-6Al-4V) during laser materials processing. It is findings like these that assist model designers in producing models that are within reasonable error in the shortest time possible.

ACKNOWLEDGMENTS

This research was sponsored under project AMTS-09-05-P, administered by the Technology Innovation Agency of South Africa.

REFERENCES

[1] Tian, Y., Chen, C. Z., Li, S. T., and Huo, Q. H., 2005, "Research Progress on Laser Surface Modification of Titanium Alloys," Applied Surface Science, **242**(1), pp. 177-184.

[2] Riabkina-Fishman, M., and Zahavi, J., 1996, "Laser Alloying and Cladding for Improving Surface Properties," Applied Surface Science, **106**, pp. 263-267.

[3] Sun, R., Yang, D. Z., Guo, L. X., and Dong, S. L., 2001, "Laser Cladding of Ti-6Al-4V Alloy with TiC and TiC+ NiCrBSi Powders," Surface and Coatings Technology, **135**(2), pp. 307-312.

[4] Saleh, A., Abboud, J. H., and Benyounis, K. Y., 2010, "Surface Carburizing of Ti-6Al-4V Alloy by Laser Melting," Optics and Lasers in Engineering, **48**(3), pp. 257-267.

[5] Pleshakov, E., Senyavs'kyi, Y., and Filip, R., 2002, "Laser Surface Modification of Ti-6Al-4V Alloy with Silicon Carbide," Materials Science, **38**(5), pp. 646-652.

[6] Rozmus-Gornikowska, M., 2010, "Surface Modifications of a Ti6Al4 V Alloy by a Laser Shock Processing," Acta Physica Polonica-Series A, **117**(5), pp. 808-811.

[7] Semak, V. V., Damkroger, B., and Kempka, S., 1999, "Temporal Evolution of the Temperature Field in the Beam Interaction Zone during Laser Material Processing," Journal of Physics D: Applied Physics, **32**(15), pp. 1819-1825.

[8] Xie, J., and Kar, A., 1997, "Mathematical Modeling of Melting during Laser Materials Processing," Journal of Applied physics, **81**(7), pp. 3015-3022.

[9] Shen, Z. H., Zhang, S. Y., Lu, J., and Ni, X.W., 2001, "Mathematical Modeling of Laser Induced Heating and Melting in Solids," Optics & Laser Technology, **33**(8), pp. 533-537.

[10] Yilbas, B. S., and Shuja, S., 1997, "Heat Transfer Analysis of Laser Heated Surfaces - Conduction Limited Case," Applied Surface Science, **108**(1), pp. 167-175.

[11] Abd El-Ghany, S. E. S., 2001, "The Temperature profile in the Molten Layer of a Semi-Infinite Target Induced by Irradiation Using a Pulsed Laser," Optics & Laser Technology, **33**(8), pp. 539-551.

[12] Tokarev, V. N., and Kaplan, A, F, H., 1999, "An analytical Modeling of Time Dependent Pulsed Laser Melting," Journal of Applied Physics, **86**(5), pp. 2836-2846.

[13] Boivineau, M., Cagran, C., Doytler, D., Eyraud, V., Nadal, M. H., Willthan, B., and Pottlacher, G., 2006, "Thermophysical Properties of Solid and Liquid Ti-6Al-4V (TA6V) Alloy," International Journal of Thermophysics, **27**(2), pp. 507-529.

Copyright © 2012 by ASME

[14] Elmer, J., Palmer, T. A., Babu, S. S., Zhang, W., and DebRoy, T., 2004, "Phase Transformation Dynamics during Welding of Ti-6Al-4V," Journal of Applied Physics, **95**(12), pp. 8327-8339.

[15] Sankaranarayanan, S., and Kar, A., 1999, "Nonlinear Effects of Laser-Plasma Interaction on Melt-Surface Temperature," Journal of Physics D: Applied Physics, **32**(7), pp. 777-784.

Copyright © 2012 by ASME

View publication stats